

Hurghada Hospital Management System (HHMS)

Project Management Plan

Course: Software Engineering Instructor: Dr. Wesam Ahmed Date: October 11, 2026

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1. Overview

We follow Scrum: thirteen one-week sprints from September 28 to December 25, 2026. New features stop on December 11, and the last two weeks are for testing, bug fixing and the final demonstration on December 25.

2. Team Organization

- **Team leader:** planning, coordination, code reviews and merging pull requests.
- **Backend (5 members):** API and database. Tasks are assigned at each sprint planning to balance the work; one member also owns the n8n automations.
- **Frontend (2 members):** app identity and all staff and patient screens.
- **Testing (2 members):** test cases, testing every feature, bug reports and progress tracking.

3. Planning

Every task is a card on our public Trello board (trello.com/b/mb3fqB6Y), moving through Backlog → Sprint → In Progress → Testing → Done. A card reaches Done only after it is tested. Each week starts with sprint planning, has short stand-ups, and ends with a review and retrospective.

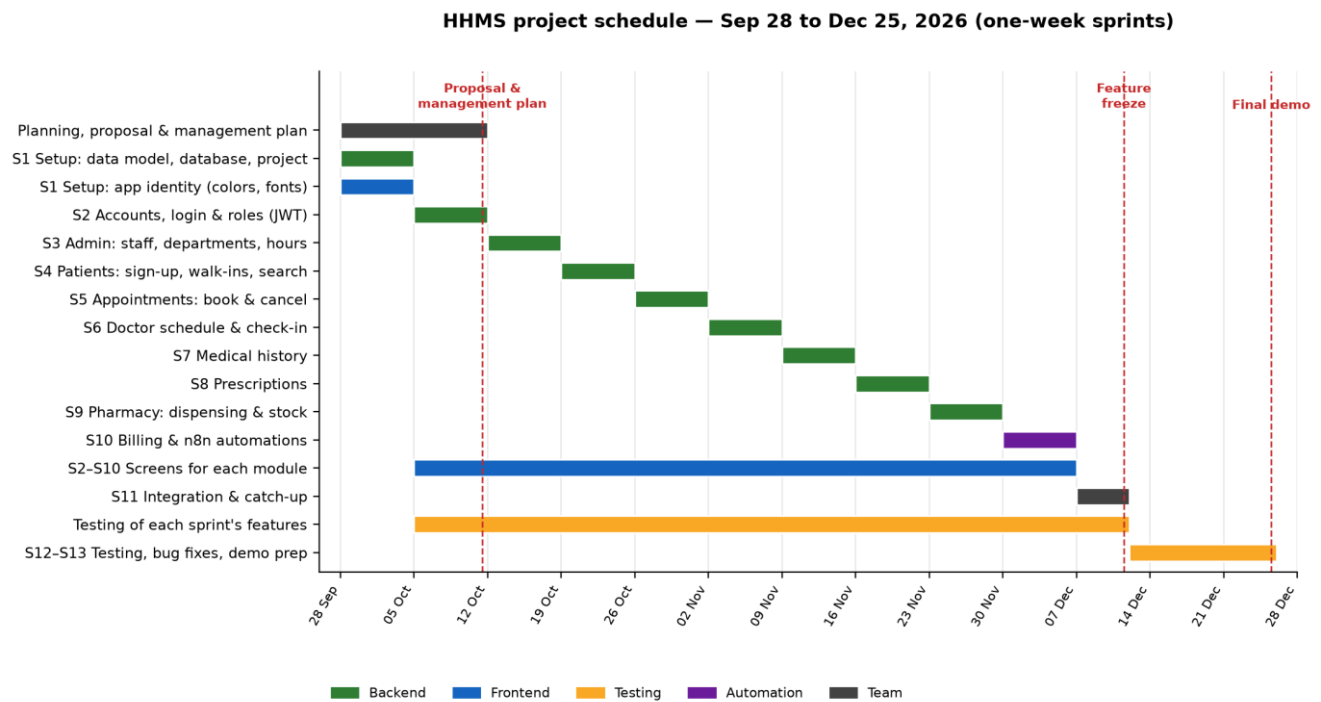
Sprint	Dates	Goal	Status
1	Sep 28 – Oct 4	Data model, database schema, project setup, app identity	Completed
2	Oct 5 – 11	Accounts, login with JWT, user roles; login and sign-up screens	In progress
3	Oct 12 – 18	Admin: staff accounts, departments, doctors' working hours	Planned
4	Oct 19 – 25	Patients: sign-up, profile, walk-in registration, search	Planned
5	Oct 26 – Nov 1	Appointments: booking and cancellation	Planned
6	Nov 2 – 8	Doctor's schedule, patient check-in	Planned
7	Nov 9 – 15	Medical history	Planned
8	Nov 16 – 22	Prescriptions	Planned
9	Nov 23 – 29	Pharmacy: dispensing and stock	Planned
10	Nov 30 – Dec 6	Billing, n8n automations, screen polish	Planned
11	Dec 7 – 11	Integration and catch-up (feature freeze on Dec 11)	Planned
12	Dec 12 – 18	Testing, bug fixes, demo script and presentation	Planned
13	Dec 19 – 25	Final tests, rehearsal, documentation, final demo	Planned

4. Communication and Workflow

Tool	Used for	How often
WhatsApp	Quick coordination and stand-up updates	Daily
Discord	Sprint planning, review and retrospective calls; pairing on hard tasks	Weekly
GitHub	Code, pull requests and code reviews	Every task
Trello	Tasks, assignments, progress and bug reports	Daily

- One branch and one pull request per task, reviewed by the team leader before merging.
- One shared SQL Server database: passwords stay in local user secrets, never in the repository; one person runs migrations; only fake data is used.

5. Scheduling



Milestones: database ready on Oct 4 (achieved) · proposal and plan on Oct 11 · feature freeze on Dec 11 · final demonstration on Dec 25.

6. Risk Management

We review these risks at every sprint retrospective.

Risk	Likelihood	Impact	How we reduce it
Inactive member or unequal workload	Medium	High	Small tasks assigned weekly; progress tracked on Trello; unfinished tasks reassigned
New technologies and frontend–backend integration	Medium	High	Setup sprint first; documented API; a dedicated integration sprint; pair work
Leaked credentials or exposed medical data	Medium	High	Secrets never committed; fake data; role checks on every request; row versioning
Merge conflicts	High	Medium	One branch per task; small pull requests; one person owns migrations
Scope creep	Medium	High	Fixed feature list; new ideas go to the backlog; freeze on Dec 11
Demo-day failure	Low	High	Two weeks of testing; rehearsal; runs locally with no deployment; backup recording

7. Quality and Documentation

- Testers test every feature before it reaches Done and report problems as Bug cards; Sprint 11 is a full end-to-end test of every role.
- The GitHub README is our live documentation (team, features, stack, database design, plan and contributions), with a separate guide to every database entity.
- A presentation and final documentation accompany the December 25 demonstration.